1. The 11th term of an arithmetic progression is 32. The sum of the first 40 terms of the arithmetic progression is 2230.

Find the 30th term of this progression. [5]

2. It is given that

$$\int_1^a 6x^5 - 6x^2 \, \mathrm{d}x = 49,$$

where $a > 1$.

(a) Show that

$$a^6 - 2a^3 - 48 = 0.$$ [4]

(b) Find the value of a. [3]

3. The graph of

$$y = x^2 + 8x + 5$$

is translated by the vector $\begin{pmatrix} p \\ q \end{pmatrix}$ to become the graph of

$$y = x^2 - 2x.$$

Find the values of the constants p and q. [4]

4. **(a)** Factorise

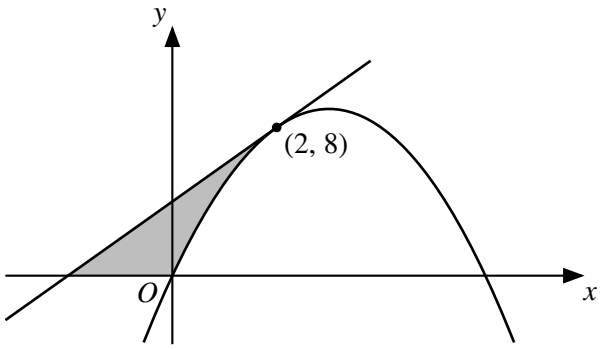
$$12a^2 - 5ab - 2b^2.$$ [1]

(b) Hence, or otherwise, solve

$$12\sin^2\theta - 5\sin\theta\cos\theta - 2\cos^2\theta = 0 \quad \text{for} \quad 0° \leqslant \theta \leqslant 360°.$$ [4]

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5.



The diagram shows a curve with equation

$$y = 9 - (x - 3)^2$$

and the tangent to the curve at the point (2, 8).

(a) State the coordinates of the maximum point on the curve. [1]

(b) Find the equation of the tangent. Give your answer in the form $y = mx + c$. [4]

(c) The shaded region is bounded by the tangent, the curve and the x-axis.

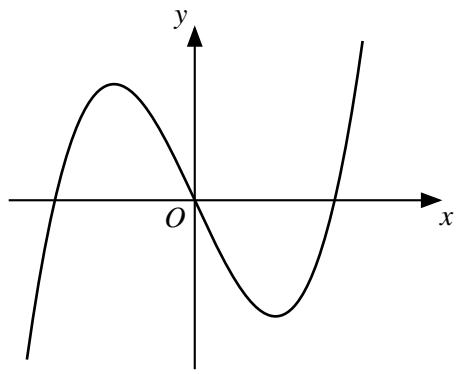
Find the exact area of the shaded region. [5]

[Turn over

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6. The function f is defined by

$$\mathrm{f}(x) = x^3 - x \qquad \text{for } x \in \mathbb{R}.$$



The diagram shows the graph of $y = \mathrm{f}(x)$.

(a) Find the set of values of x for which f is decreasing. [3]

(b) The equation $\mathrm{f}(x) = k$, where k is a constant, has two distinct solutions.

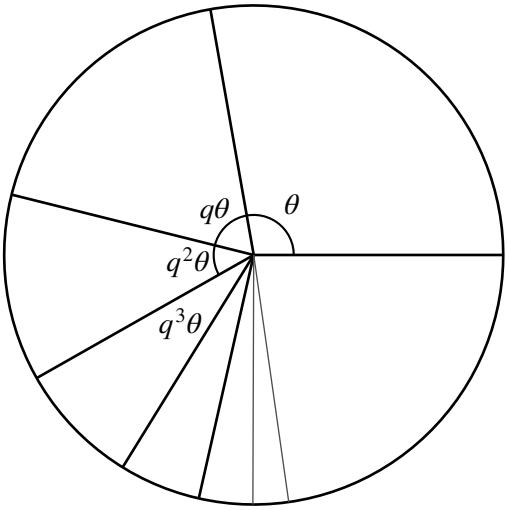
Find the possible values of k. [2]

(c) The graph of $y = \mathrm{f}(x)$ is stretched by scale factor $\frac{1}{2}$ in the x-direction and then translated by the vector $\begin{pmatrix} 0 \\ 7 \end{pmatrix}$.

Find the equation of the transformed curve. [2]

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7.



The diagram shows a sequence of sectors contained within the same circle of radius r cm. The angles made at the centre of the circle in consecutive sectors form a geometric progression with first term θ radians and common ratio q, where q is a constant such that $0 < q < 1$.

(a) Find an expression for the area of the nth sector in the sequence. Give your answer in terms of q, θ, r and n. [1]

(b) It is given that as the number of sectors tends to infinity, the sectors completely fill the circle without overlapping.

Show that

$$q = 1 - \frac{\theta}{2\pi}.$$ [3]

(c) It is now given that $\theta = \frac{3}{5}\pi$ and $r = 15$.

Find the sum of the arc lengths of the first 10 sectors in the sequence. [3]

8. It is given that

$$\tan\alpha = k \qquad \text{where } 180° < \alpha < 270°.$$

Find expressions, in terms of k, for the exact values of $\sin\alpha$ and $\cos\alpha$. [4]

9. **(a)** Find the coefficient of x in the expansion of

$$\left(x^2 - \frac{2}{x^3}\right)^3.$$ [2]

(b) Find the term independent of x in the expansion of

$$\left(x^2 - \frac{2}{x^3}\right)^{15}.$$ [3]

[Turn over

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10. A circle has equation

$$x^2 + y^2 = 9.$$

The line with equation $y = mx + c$ is a tangent to the circle, where m and c are positive constants.

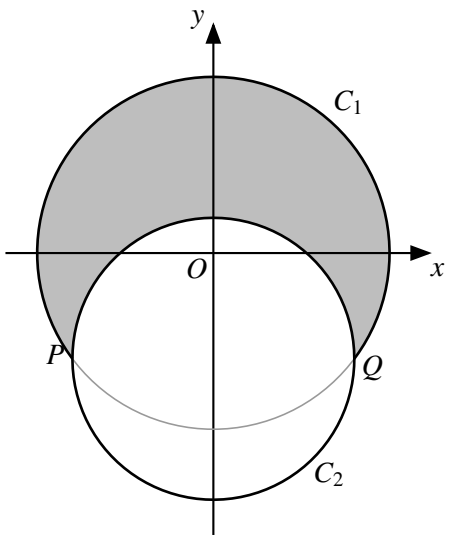
(a) Show that

$$c^2 = 9(m^2 + 1).$$ [4]

(b) The tangent intersects the x-axis at A and the y-axis at B. The area of triangle AOB is 15 square units.

Find all possible equations of the tangent. [7]

11.



The diagram shows a design for a logo. The design consists of two circles C_1 and C_2 with equations given by

$$C_1\colon\ x^2 + y^2 = 100$$

and

$$C_2\colon\ x^2 + (y + 6)^2 = 64.$$

The two circles intersect at the points P and Q, as shown on the diagram.

(a) Show that the line segment PQ is a diameter of C_2. [3]

(b) Find, in radians, the angle POQ. Give your answer correct to 3 decimal places. [3]

(c) The shaded region is inside C_1 but outside C_2.

Find the area of the shaded region. [4]

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